

Mandatory: Supervised Learning I

ITF31519-1 25H Praktisk maskinlæring

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September 21, 2025
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Classification

1.1 Task

Create a decision tree manually for [this dataset](#).

You must use either entropy or Gini impurity to find the best splits.

The dataset is a subset of the Iris dataset. Use the first 12 rows for training and the three remaining instances for testing. The dataset contains numerical features only, so the procedure will be slightly different from what we've covered in the lectures. The covered algorithm (ID3) must be combined with a procedure for numerical features.

In ID3, we find the next feature by finding the feature with highest information gain. The information gain calculations must be adjusted so that the best threshold to split the feature is found i.e. each feature will have a separate information gain for each potential threshold.

You must show all relevant calculations. You can skip calculations if you can provide logical reasoning can replace them. Show the decision tree, both as it grows and the final version.

What is the accuracy of the decision tree on the test instances? Discuss the results.

1.2 Solution

1.2.1 Selecting the data

Since the task specify that the first 12 rows shall be used for training and the rest for testing. We end up with the following partitioning of the given dataset.

number	sepal_length(cm)"	"sepal_width(cm)"	"petal_length(cm)"	"petal_width(cm)"	"class"	
0	4.4	3.2	1.3	0.2	Iris-setosa	Train
1	4.9	3.1	1.5	0.1	Iris-setosa	
2	4.6	3.2	1.4	0.2	Iris-setosa	
3	4.4	2.9	1.4	0.2	Iris-setosa	
4	6	2.7	5.1	1.6	Iris-versicolor	
5	5.8	2.7	3.9	1.2	Iris-versicolor	
6	6.1	2.9	4.7	1.4	Iris-versicolor	
7	5.7	3	4.2	1.2	Iris-versicolor	
8	6.7	2.5	5.8	1.8	Iris-virginica	
9	6.8	3.2	5.9	2.3	Iris-virginica	
10	6.8	3	5.5	2.1	Iris-virginica	
11	6.3	3.4	5.6	2.4	Iris-virginica	
12	4.6	3.2	1.4	0.2	Iris-setosa	Test
13	5.1	2.5	3	1.1	Iris-versicolor	
14	7.2	3	5.8	1.6	Iris-virginica	

Figure 1.1: Screenshot from excel formatting of the dataset.

1.2.2 Entropy

1.2.2.1 General formula

- Entropy

$$H(S) = - \sum_{i=1}^k p_i \log_2(p_i)$$

Where:

- S = Entire dataset.
- k = Unique classes in the target column.
- $p_i = \frac{\text{count of class } i}{\text{total samples}}$
- Information Gain

$$IG(S, A) = H(S) - \sum_{t \in T} p(t)H(t) = H(S) - H(S | A)$$

Where T are the subsets created by splitting on A .

1.2.2.2 Finding the Root Node

Now that the dataset has been divided into train/test partitions, we can start to calculate the 'Root Node' of the training set.

In this case the target column have three possible outcomes:

- Iris-setosa $p(\frac{1}{3})$
- Iris-versicolor $p(\frac{1}{3})$
- Iris-virginica $p(\frac{1}{3})$

This gives us the following formula for the 'Root Node':

$$H(S) = - \sum_{i=1}^3 p_i \log_2(p_i)$$

$$H(S) = -(\frac{1}{3} \log_2(\frac{1}{3}) + \frac{1}{3} \log_2(\frac{1}{3}) + \frac{1}{3} \log_2(\frac{1}{3}))$$

Applying the log rule: $\log_a(\frac{x}{y}) = \log_a(x) - \log_a(y)$:

$$H(S) = -(\frac{1}{3} \log_2(1) - \log_2(3) + \frac{1}{3} \log_2(1) - \log_2(3) + \frac{1}{3} \log_2(1) - \log_2(3))$$

$$\log_2(1) = 0 \quad \log_2(3) \approx 1.58496$$

$$\begin{aligned}
H(S) &= -\left(\frac{1}{3} \cdot (-1.58496) + \frac{1}{3} \cdot (-1.58496) + \frac{1}{3} \cdot (-1.58496)\right) \\
H(S) &= -\left(\left(-\frac{1.58496}{3}\right) - \frac{1.58496}{3} - \frac{1.58496}{3}\right) \\
H(S) &= -\left(\frac{-(1.58496) - 1.58496 - 1.58496}{3}\right) \\
H(S) &= -\left(-\frac{4.75488}{3}\right) \\
H(S) &= -(-1.58496) \\
\underline{\underline{H(S) = 1.58496}}
\end{aligned}$$

1.2.2.3 Calculating the Information Gain(IG)

Now that we've found the *Entropy* for the *Root Node*, we can start calculating the *IG* for the features in the training dataset.

We'll start with the first feature from the left and work our way through all the feature columns in the training dataset("sepal_length(cm)" then "sepal_width(cm)" then "petal_length(cm)" then "petal_width(cm)").

"sepal_length(cm)"	"sepal_width(cm)"	"petal_length(cm)"	"petal_width(cm)"	"class"
4.4	3.2	1.3	0.2	Iris-setosa
4.9	3.1	1.5	0.1	Iris-setosa
4.6	3.2	1.4	0.2	Iris-setosa
4.4	2.9	1.4	0.2	Iris-setosa
6	2.7	5.1	1.6	Iris-versicolor
5.8	2.7	3.9	1.2	Iris-versicolor
6.1	2.9	4.7	1.4	Iris-versicolor
5.7	3	4.2	1.2	Iris-versicolor
6.7	2.5	5.8	1.8	Iris-virginica
6.8	3.2	5.9	2.3	Iris-virginica
6.8	3	5.5	2.1	Iris-virginica
6.3	3.4	5.6	2.4	Iris-virginica

Table 1.1: Train features

1.2.2.4 sepal_length(cm)

This feature consist of the following values:

Value	Instances
4.4	2
4.6	1
4.9	1
5.7	1
5.8	1
6	1
6.1	1
6.3	1
6.7	1
6.8	2

Table 1.2: Instance count

With this information we can now calculate the *entropy* for this feature.

$$H(S_{sepal_length(cm)=4.4}) \rightarrow -\left(\frac{2}{2} \log_2 \frac{2}{2} + \frac{0}{2} \log_2 \frac{0}{2} + \frac{0}{2} \log_2 \frac{0}{2}\right) \rightarrow 0$$

$$H(S_{sepal_length(cm)=4.6}) \rightarrow -\left(\frac{1}{1} \log_2 \frac{1}{1} + \frac{0}{1} \log_2 \frac{0}{1} + \frac{0}{1} \log_2 \frac{0}{1}\right) \rightarrow 0$$

$$H(S_{sepal_length(cm)=4.9}) \rightarrow -\left(\frac{1}{1} \log_2 \frac{1}{1} + \frac{0}{1} \log_2 \frac{0}{1} + \frac{0}{1} \log_2 \frac{0}{1}\right) \rightarrow 0$$

$$H(S_{sepal_length(cm)=5.7}) \rightarrow -\left(\frac{0}{1} \log_2 \frac{0}{1} + \frac{1}{1} \log_2 \frac{1}{1} + \frac{0}{1} \log_2 \frac{0}{1}\right) \rightarrow 0$$

$$H(S_{sepal_length(cm)=5.8}) \rightarrow -\left(\frac{0}{1} \log_2 \frac{0}{1} + \frac{1}{1} \log_2 \frac{1}{1} + \frac{0}{1} \log_2 \frac{0}{1}\right) \rightarrow 0$$

$$H(S_{sepal_length(cm)=6}) \rightarrow -\left(\frac{0}{1} \log_2 \frac{0}{1} + \frac{1}{1} \log_2 \frac{1}{1} + \frac{0}{1} \log_2 \frac{0}{1}\right) \rightarrow 0$$

$$H(S_{sepal_length(cm)=6.1}) \rightarrow -\left(\frac{0}{1} \log_2 \frac{0}{1} + \frac{1}{1} \log_2 \frac{1}{1} + \frac{0}{1} \log_2 \frac{0}{1}\right) \rightarrow 0$$

$$H(S_{sepal_length(cm)=6.3}) \rightarrow -\left(\frac{0}{1} \log_2 \frac{0}{1} + \frac{0}{1} \log_2 \frac{0}{1} + \frac{1}{1} \log_2 \frac{1}{1}\right) \rightarrow 0$$

$$H(S_{sepal_length(cm)=6.7}) \rightarrow -\left(\frac{0}{1} \log_2 \frac{0}{1} + \frac{0}{1} \log_2 \frac{0}{1} + \frac{1}{1} \log_2 \frac{1}{1}\right) \rightarrow 0$$

$$H(S_{sepal_length(cm)=6.8}) \rightarrow -\left(\frac{0}{2} \log_2 \frac{0}{2} + \frac{0}{2} \log_2 \frac{0}{2} + \frac{2}{2} \log_2 \frac{2}{2}\right) \rightarrow 0$$

With this information we can now calculate $IG(S, sepal_length(cm))$

$$IG(S, sepal_length(cm)) = H(S) - H(S | A)$$

$$1.58496 - \frac{1}{6} \cdot 0 - \frac{1}{12} \cdot 0 - \frac{1}{12} \cdot 0 - \frac{1}{12} \cdot 0 - \frac{1}{12} \cdot 0 - \frac{1}{12} \cdot 0 - \frac{1}{12} \cdot 0 - \frac{1}{12} \cdot 0 - \frac{1}{12} \cdot 0 - \frac{1}{6} \cdot 0$$

$$\underline{IG(S, sepal_length(cm)) \approx 1.58496}$$

1.2.2.5 sepal_width(cm)

This feature consist of the following values:

Value	Instances
2.5	1
2.7	2
2.9	2
3	2
3.1	1
3.2	3
3.4	1

Table 1.3: Instance count

With this information we can now calculate the *entropy* for this feature.

$$H(S_{sepal_width(cm)=2.5}) \rightarrow -\left(\frac{0}{1} \log_2 \frac{0}{1} + \frac{0}{1} \log_2 \frac{0}{1} + \frac{1}{1} \log_2 \frac{1}{1}\right) \rightarrow 0$$

$$H(S_{sepal_width(cm)=2.7}) \rightarrow -\left(\frac{0}{2} \log_2 \frac{0}{2} + \frac{2}{2} \log_2 \frac{2}{2} + \frac{0}{2} \log_2 \frac{0}{2}\right) \rightarrow 0$$

$$H(S_{sepal_width(cm)=2.9}) \rightarrow -\left(\frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{2} \log_2 \frac{1}{2} + \frac{0}{2} \log_2 \frac{0}{2}\right) \rightarrow 1$$

$$H(S_{sepal_width(cm)=3}) \rightarrow -\left(\frac{0}{2} \log_2 \frac{0}{2} + \frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{2} \log_2 \frac{1}{2}\right) \rightarrow 1$$

$$H(S_{sepal_width(cm)=3.1}) \rightarrow -\left(\frac{1}{1} \log_2 \frac{1}{1} + \frac{0}{1} \log_2 \frac{0}{1} + \frac{0}{1} \log_2 \frac{0}{1}\right) \rightarrow 0$$

$$H(S_{sepal_width(cm)=3.2}) \rightarrow -\left(\frac{2}{3} \log_2 \frac{2}{3} + \frac{0}{3} \log_2 \frac{0}{3} + \frac{1}{3} \log_2 \frac{1}{3}\right) \rightarrow 0.918291$$

$$H(S_{sepal_width(cm)=3.4}) \rightarrow -\left(\frac{0}{1} \log_2 \frac{0}{1} + \frac{0}{1} \log_2 \frac{0}{1} + \frac{1}{1} \log_2 \frac{1}{1}\right) \rightarrow 0$$

With this information we can now calculate $IG(S, sepal_width(cm))$

$$\begin{aligned}
 IG(S, sepal_width(cm)) &= H(S) - H(S | A) \\
 1.58496 - \frac{1}{12} \cdot 0 - \frac{1}{6} \cdot 0 - \frac{1}{6} \cdot 1 - \frac{1}{6} \cdot 1 - \frac{1}{12} \cdot 0 - \frac{1}{4} \cdot 0.918291 - \frac{1}{12} \cdot 0 \\
 \underline{IG(S, sepal_width(cm)) \approx 1.02205}
 \end{aligned}$$

1.2.2.6 petal_length(cm)

This feature consist of the following values:

Value	Instances
1.3	1
1.4	2
1.5	1
3.9	1
4.2	1
4.7	1
5.1	1
5.5	1
5.6	1
5.8	1
5.9	1

Table 1.4: Instance count

With this information we can now calculate the *entropy* for this feature.

$$H(S_{petal_length(cm)=1.3}) \rightarrow -\left(\frac{1}{1} \log_2 \frac{1}{1} + \frac{0}{1} \log_2 \frac{0}{1} + \frac{0}{1} \log_2 \frac{0}{1}\right) \rightarrow 0$$

$$H(S_{petal_length(cm)=1.4}) \rightarrow -\left(\frac{2}{2} \log_2 \frac{2}{2} + \frac{0}{2} \log_2 \frac{0}{2} + \frac{0}{2} \log_2 \frac{0}{2}\right) \rightarrow 0$$

$$H(S_{petal_length(cm)=1.5}) \rightarrow -\left(\frac{1}{1} \log_2 \frac{1}{1} + \frac{0}{1} \log_2 \frac{0}{1} + \frac{0}{1} \log_2 \frac{0}{1}\right) \rightarrow 0$$

$$H(S_{petal_length(cm)=3.9}) \rightarrow -\left(\frac{0}{1} \log_2 \frac{0}{1} + \frac{1}{1} \log_2 \frac{1}{1} + \frac{0}{1} \log_2 \frac{0}{1}\right) \rightarrow 0$$

$$H(S_{petal_length(cm)=4.2}) \rightarrow -\left(\frac{0}{1} \log_2 \frac{0}{1} + \frac{1}{1} \log_2 \frac{1}{1} + \frac{0}{1} \log_2 \frac{0}{1}\right) \rightarrow 0$$

$$H(S_{petal_length(cm)=4.7}) \rightarrow -\left(\frac{0}{1} \log_2 \frac{0}{1} + \frac{1}{1} \log_2 \frac{1}{1} + \frac{0}{1} \log_2 \frac{0}{1}\right) \rightarrow 0$$

$$H(S_{petal_length(cm)=5.1}) \rightarrow -\left(\frac{0}{1} \log_2 \frac{0}{1} + \frac{1}{1} \log_2 \frac{1}{1} + \frac{0}{1} \log_2 \frac{0}{1}\right) \rightarrow 0$$

$$H(S_{petal_length(cm)=5.5}) \rightarrow -\left(\frac{0}{1} \log_2 \frac{0}{1} + \frac{0}{1} \log_2 \frac{0}{1} + \frac{1}{1} \log_2 \frac{1}{1}\right) \rightarrow 0$$

$$H(S_{petal_length(cm)=5.6}) \rightarrow -\left(\frac{0}{1} \log_2 \frac{0}{1} + \frac{0}{1} \log_2 \frac{0}{1} + \frac{1}{1} \log_2 \frac{1}{1}\right) \rightarrow 0$$

$$H(S_{petal_length(cm)=5.8}) \rightarrow -\left(\frac{0}{1} \log_2 \frac{0}{1} + \frac{0}{1} \log_2 \frac{0}{1} + \frac{1}{1} \log_2 \frac{1}{1}\right) \rightarrow 0$$

$$H(S_{petal_length(cm)=5.9}) \rightarrow -\left(\frac{0}{1} \log_2 \frac{0}{1} + \frac{0}{1} \log_2 \frac{0}{1} + \frac{1}{1} \log_2 \frac{1}{1}\right) \rightarrow 0$$

With this information we can now calculate $IG(S, petal_length(cm))$

$$IG(S, petal_length(cm)) = H(S) - H(S | A)$$

$$1.58496 - \frac{1}{12} \cdot 0 - \frac{1}{6} \cdot 0 - \frac{1}{12} \cdot 0 - \frac{1}{12} \cdot 0 - \frac{1}{12} \cdot 0 - \frac{1}{12} \cdot 0 - \frac{1}{12} \cdot 0 - \frac{1}{12} \cdot 0 - \frac{1}{12} \cdot 0 - \frac{1}{12} \cdot 0 - \frac{1}{12} \cdot 0$$

$$\underline{IG(S, petal_length(cm)) \approx 1.58496}$$

1.2.2.7 petal_width(cm)

This feature consist of the following values:

Value	Instances
0.1	1
0.2	3
1.2	2
1.4	1
1.6	1
1.8	1
2.1	1
2.3	1
2.4	1

Table 1.5: Instance count

With this information we can now calculate the *entropy* for this feature.

$$H(S_{petal_width(cm)=0.1}) \rightarrow -\left(\frac{1}{1} \log_2 \frac{1}{1} + \frac{0}{1} \log_2 \frac{0}{1} + \frac{0}{1} \log_2 \frac{0}{1}\right) \rightarrow 0$$

$$H(S_{petal_width(cm)=0.2}) \rightarrow -\left(\frac{3}{3} \log_2 \frac{3}{3} + \frac{0}{3} \log_2 \frac{0}{3} + \frac{0}{3} \log_2 \frac{0}{3}\right) \rightarrow 0$$

$$H(S_{petal_width(cm)=1.2}) \rightarrow -\left(\frac{0}{2} \log_2 \frac{0}{2} + \frac{2}{2} \log_2 \frac{2}{2} + \frac{0}{2} \log_2 \frac{0}{2}\right) \rightarrow 0$$

$$H(S_{petal_width(cm)=1.4}) \rightarrow -\left(\frac{0}{1} \log_2 \frac{0}{1} + \frac{1}{1} \log_2 \frac{1}{1} + \frac{0}{1} \log_2 \frac{0}{1}\right) \rightarrow 0$$

$$H(S_{petal_width(cm)=1.6}) \rightarrow -\left(\frac{0}{1} \log_2 \frac{0}{1} + \frac{1}{1} \log_2 \frac{1}{1} + \frac{0}{1} \log_2 \frac{0}{1}\right) \rightarrow 0$$

$$H(S_{petal_width(cm)=1.8}) \rightarrow -\left(\frac{0}{1} \log_2 \frac{0}{1} + \frac{0}{1} \log_2 \frac{0}{1} + \frac{1}{1} \log_2 \frac{1}{1}\right) \rightarrow 0$$

$$H(S_{petal_width(cm)=2.1}) \rightarrow -\left(\frac{0}{1} \log_2 \frac{0}{1} + \frac{0}{1} \log_2 \frac{0}{1} + \frac{1}{1} \log_2 \frac{1}{1}\right) \rightarrow 0$$

$$H(S_{petal_width(cm)=2.3}) \rightarrow -\left(\frac{0}{1} \log_2 \frac{0}{1} + \frac{0}{1} \log_2 \frac{0}{1} + \frac{1}{1} \log_2 \frac{1}{1}\right) \rightarrow 0$$

$$H(S_{petal_width(cm)=2.4}) \rightarrow -\left(\frac{0}{1} \log_2 \frac{0}{1} + \frac{0}{1} \log_2 \frac{0}{1} + \frac{1}{1} \log_2 \frac{1}{1}\right) \rightarrow 0$$

With this information we can now calculate $IG(S, petal_width(cm))$

$$IG(S, petal_width(cm)) = H(S) - H(S | A)$$

$$1.58496 - \frac{1}{12} \cdot 0 - \frac{1}{4} \cdot 0 - \frac{1}{6} \cdot 0 - \frac{1}{12} \cdot 0 - \frac{1}{12} \cdot 0 - \frac{1}{12} \cdot 0 - \frac{1}{12} \cdot 0 - \frac{1}{12} \cdot 0 - \frac{1}{12} \cdot 0$$

$$\underline{IG(S, petal_width(cm)) \approx 1.58496}$$

1.2.2.8 Choosing the first node

Summarizing results

- $IG(S, sepal_length(cm)) \approx 1.58496$
- $IG(S, sepal_width(cm)) \approx 1.02205$
- $IG(S, petal_length(cm)) \approx 1.58496$
- $IG(S, petal_width(cm)) \approx 1.58496$

Here we observe that three features ends up with having the highest *IG score*. To solve this I'll try to calculate a new *IG score*, but with a **binary threshold**.

1.2.2.9 Recompute IG with Binary Thresholds

Starting at the top of the list and work our way through.

Firstly the unique values of the given feature is sorted in a list. Then the θ values is calculated for each list with the following formula:

Information Gain for a threshold split on A at θ . Let S be the current node with $|S| = N$ and class set $\mathcal{Y} = \{1, \dots, K\}$. Let $n_i = \#\{(x, y) \in S : y = i\}$ and define

$$H(S) = - \sum_{i=1}^K \frac{n_i}{N} \log_2 \left(\frac{n_i}{N} \right) \quad (\text{with } 0 \log 0 := 0).$$

For a threshold θ on numeric feature A ,

$$S_{\leq \theta} = \{x \in S : A(x) \leq \theta\}, \quad S_{> \theta} = \{x \in S : A(x) > \theta\}.$$

Let $N_L = |S_{\leq \theta}|$, $N_R = |S_{> \theta}|$ and

$$n_{L,i} = \#\{(x, y) \in S_{\leq \theta} : y = i\}, \quad n_{R,i} = \#\{(x, y) \in S_{> \theta} : y = i\}.$$

Subset entropies:

$$H(S_{\leq \theta}) = - \sum_{i=1}^K \frac{n_{L,i}}{N_L} \log_2 \left(\frac{n_{L,i}}{N_L} \right), \quad H(S_{> \theta}) = - \sum_{i=1}^K \frac{n_{R,i}}{N_R} \log_2 \left(\frac{n_{R,i}}{N_R} \right).$$

Conditional entropy for the split (A, θ) :

$$H(S \mid A, \theta) = \frac{N_L}{N} H(S_{\leq \theta}) + \frac{N_R}{N} H(S_{> \theta}).$$

Information gain:

$$IG(S, A, \theta) = H(S) - H(S \mid A, \theta).$$

Candidate thresholds. For the sorted distinct values $U_1 < \dots < U_m$ of A in S ,

$$\Theta(A) = \left\{ \theta_j = \frac{U_j + U_{j+1}}{2} : j = 1, \dots, m-1 \right\}.$$

Pick the best threshold for A :

$$\theta_A^* \in \arg \max_{\theta \in \Theta(A)} IG(S, A, \theta),$$

and the root split as

$$(A^*, \theta^*) \in \arg \max_A IG(S, A, \theta_A^*).$$

· **S,sepal_length(cm)**

$$U = \{4.4, 4.6, 4.9, 5.7, 5.8, 6, 6.1, 6.3, 6.7, 6.8\}$$

$$\theta \in \left\{ \frac{4.4 + 4.6}{2}, \frac{4.6 + 4.9}{2}, \frac{4.9 + 5.7}{2}, \frac{5.7 + 5.8}{2}, \frac{5.8 + 6}{2}, \frac{6 + 6.1}{2}, \frac{6.1 + 6.3}{2}, \frac{6.3 + 6.7}{2}, \frac{6.7 + 6.8}{2} \right\}$$

$$\theta \in \{4.5, 4.75, 5.3, 5.75, 5.9, 6.05, 6.2, 6.5, 6.75\}$$

$$\begin{aligned}
S_{\leq 4.50} &= \{x : \text{sepal_length}(x) \leq 4.50\}, \quad S_{>4.50} = \{x : \text{sepal_length}(x) > 4.5\} \\
|S_{\leq 4.50}| &= 2, \text{ class counts } (2, 0, 0), \quad |S_{>4.50}| = 10, \text{ class counts } (2, 4, 4) \\
H(S_{\leq 4.50}) &= -(1 \log_2 1 + 0 \log_2 0 + 0 \log_2 0) = 0 \\
H(S_{>4.50}) &= -\left(\frac{1}{5} \log_2 \frac{1}{5} + \frac{2}{5} \log_2 \frac{2}{5} + \frac{2}{5} \log_2 \frac{2}{5}\right) \approx 1.52192 \\
H(S \mid \text{sepal_length}, 4.50) &= \frac{2}{12} \cdot 0 + \frac{10}{12} \cdot 1.52192 = 1.26826 \\
IG(S, \text{sepal_length}, 4.50) &= 1.58496 - 1.26826 = 0.3167
\end{aligned}$$

θ	$ S_{\leq \theta} $	Left counts	H_L	$ S_{> \theta} $	Right counts	H_R	$H(S \mid A, \theta)$	IG
4.50	2	(2,0,0)	0.000000	10	(2,4,4)	1.521928	1.268273	0.316689
4.75	3	(3,0,0)	0.000000	9	(1,4,4)	1.392147	1.044110	0.540852
5.30	4	(4,0,0)	0.000000	8	(0,4,4)	1.000000	0.666667	0.918296
5.75	5	(4,1,0)	0.721928	7	(0,3,4)	0.985228	0.875520	0.709443
5.90	6	(4,2,0)	0.918296	6	(0,2,4)	0.918296	0.918296	0.666667
6.05	7	(4,3,0)	0.985228	5	(0,1,4)	0.721928	0.875520	0.709443
6.20	8	(4,4,0)	1.000000	4	(0,0,4)	0.000000	0.666667	0.918296
6.50	9	(4,4,1)	1.392147	3	(0,0,3)	0.000000	1.044110	0.540852
6.75	10	(4,4,2)	1.521928	2	(0,0,2)	0.000000	1.268273	0.316689

· **S,petal_length(cm)**

$$U = \{1.3, 1.4, 1.5, 3.9, 4.2, 4.7, 5.1, 5.5, 5.6, 5.8, 5.9\}$$

$$\begin{aligned}
\theta \in \left\{ \frac{1.3+1.4}{2}, \frac{1.4+1.5}{2}, \frac{1.5+3.9}{2}, \frac{3.9+4.2}{2}, \frac{4.2+4.7}{2}, \frac{4.7+5.1}{2}, \right. \\
\left. \frac{5.1+5.5}{2}, \frac{5.5+5.6}{2}, \frac{5.6+5.8}{2}, \frac{5.8+5.9}{2} \right\}
\end{aligned}$$

$$\theta \in \{1.35, 1.45, 2.7, 4.05, 4.45, 4.9, 5.3, 5.55, 5.69, 5.85\}$$

θ	$ S_{\leq \theta} $	Left counts	H_L	$ S_{> \theta} $	Right counts	H_R	$H(S A, \theta)$	IG
1.35	1	(1, 0, 0)	0.000000	11	(3, 4, 4)	1.572624	1.441572	0.143391
1.45	3	(3, 0, 0)	0.000000	9	(1, 4, 4)	1.392147	1.044110	0.540852
2.70	4	(4, 0, 0)	0.000000	8	(0, 4, 4)	1.000000	0.666667	0.918296
4.05	5	(4, 1, 0)	0.721928	7	(0, 3, 4)	0.985228	0.875520	0.709443
4.45	6	(4, 2, 0)	0.918296	6	(0, 2, 4)	0.918296	0.918296	0.666667
4.90	7	(4, 3, 0)	0.985228	5	(0, 1, 4)	0.721928	0.875520	0.709443
5.30	8	(4, 4, 0)	1.000000	4	(0, 0, 4)	0.000000	0.666667	0.918296
5.55	9	(4, 4, 1)	1.392147	3	(0, 0, 3)	0.000000	1.044110	0.540852
5.70	10	(4, 4, 2)	1.521928	2	(0, 0, 2)	0.000000	1.268273	0.316689
5.85	11	(4, 4, 3)	1.572624	1	(0, 0, 1)	0.000000	1.441572	0.143391

· **S,petal_width(cm)**

$$U = \{0.1, 0.2, 1.2, 1.4, 1.6, 1.8, 2.1, 2.3, 2.4\}$$

$$\theta \in \left\{ \frac{0.1 + 0.2}{2}, \frac{0.2 + 1.2}{2}, \frac{1.2 + 1.4}{2}, \frac{1.4 + 1.6}{2}, \frac{1.6 + 1.8}{2}, \frac{1.8 + 2.1}{2}, \frac{2.1 + 2.3}{2}, \frac{2.3 + 2.4}{2} \right\}$$

$$\theta \in \{0.15, 0.70, 1.30, 1.50, 1.70, 1.95, 2.20, 2.35\}$$

θ	$ S_{\leq \theta} $	Left counts	H_L	$ S_{> \theta} $	Right counts	H_R	$H(S A, \theta)$	IG
0.15	1	(1, 0, 0)	0.000000	11	(3, 4, 4)	1.572624	1.441572	0.143391
0.70	4	(4, 0, 0)	0.000000	8	(0, 4, 4)	1.000000	0.666667	0.918296
1.30	6	(4, 2, 0)	0.918296	6	(0, 2, 4)	0.918296	0.918296	0.666667
1.50	7	(4, 3, 0)	0.985228	5	(0, 1, 4)	0.721928	0.875520	0.709443
1.70	8	(4, 4, 0)	1.000000	4	(0, 0, 4)	0.000000	0.666667	0.918296
1.95	9	(4, 4, 1)	1.392147	3	(0, 0, 3)	0.000000	1.044110	0.540852
2.20	10	(4, 4, 2)	1.521928	2	(0, 0, 2)	0.000000	1.268273	0.316689
2.35	11	(4, 4, 3)	1.572624	1	(0, 0, 1)	0.000000	1.441572	0.143391

1.2.2.10 Best Candidates after Threshold Split

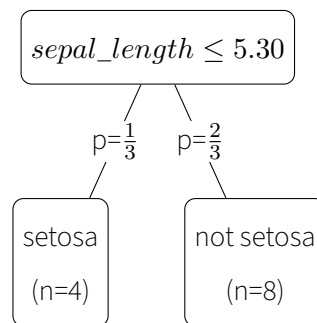
After calculating IG with binary threshold we're left with the following candidates:

- $IG(S, sepal_length, 5.30) = 0.918296$
- $IG(S, sepal_length, 6.20) = 0.918296$

- $IG(S, \text{petal_length}, 2.70) = 0.918296$
- $IG(S, \text{petal_length}, 5.30) = 0.918296$
- $IG(S, \text{petal_width}, 1.70) = 0.918296$
- $IG(S, \text{petal_width}, 0.70) = 0.918296$

All these alternatives will result in a pure leaf node on the first branch, so all these are equally qualified for the root node.

Here I choose $IG(S, \text{sepal_length}, 5.30) = 0.918296$ as the root node which results in the following tree structure



With $\text{sepal_length} \leq 5.30$ as the root node, this gives us the following remaining features:

number	sepal_length(cm)	sepal_width(cm)	petal_length(cm)	petal_width(cm)	class	
0	4.4	3.2	1.3	0.2	Iris-setosa	Solved
3	4.4	2.9	1.4	0.2	Iris-setosa	
2	4.6	3.2	1.4	0.2	Iris-setosa	
1	4.9	3.1	1.5	0.1	Iris-setosa	
7	5.7	3	4.2	1.2	Iris-versicolor	Remaining
5	5.8	2.7	3.9	1.2	Iris-versicolor	
4	6	2.7	5.1	1.6	Iris-versicolor	
6	6.1	2.9	4.7	1.4	Iris-versicolor	
11	6.3	3.4	5.6	2.4	Iris-virginica	
8	6.7	2.5	5.8	1.8	Iris-virginica	
9	6.8	3.2	5.9	2.3	Iris-virginica	
10	6.8	3	5.5	2.1	Iris-virginica	

Figure 1.2: Screenshot from excel formatting of the dataset.

1.2.2.11 $\text{sepal_length} \leq 5.30$ – Child Node

We now recalculate the next potential node based on the entropy of $\text{sepal_length} > 5.30$

$$H(S_{\text{sepal_length} > 5.30}) = -\left(0 + \frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{2} \log_2 \frac{1}{2}\right) = 1$$

We now calculate the IG for the features again, but we remove the four rows that's being sorted in the root node.

1	number	sepal_length(cm)	"sepal_width(cm)"	"petal_length(cm)"	"petal_width(cm)"	"class"	
6	7	5.7	3	4.2	1.2	Iris-versicolor	Remaining
7	5	5.8	2.7	3.9	1.2	Iris-versicolor	
8	4	6	2.7	5.1	1.6	Iris-versicolor	
9	6	6.1	2.9	4.7	1.4	Iris-versicolor	
10	11	6.3	3.4	5.6	2.4	Iris-virginica	
11	8	6.7	2.5	5.8	1.8	Iris-virginica	
12	9	6.8	3.2	5.9	2.3	Iris-virginica	
13	10	6.8	3	5.5	2.1	Iris-virginica	

Figure 1.3: Screenshot from excel formatting of the dataset.

· **sepal_length(cm)**

θ	$ S_{\leq \theta} $	Left counts	H_L	$ S_{> \theta} $	Right counts	H_R	$H(S A, \theta)$	IG
5.75	1	(1, 0)	0.000000	7	(3, 4)	0.985228	0.862075	0.137925
5.90	2	(2, 0)	0.000000	6	(2, 4)	0.918296	0.688722	0.311278
6.05	3	(3, 0)	0.000000	5	(1, 4)	0.721928	0.451205	0.548795
6.20	4	(4, 0)	0.000000	4	(0, 4)	0.000000	0.000000	1.000000
6.50	5	(4, 1)	0.721928	3	(0, 3)	0.000000	0.451205	0.548795
6.75	6	(4, 2)	0.918296	2	(0, 2)	0.000000	0.688722	0.311278

· **sepal_width(cm)**

θ	$ S_{\leq \theta} $	Left counts	H_L	$ S_{> \theta} $	Right counts	H_R	$H(S A, \theta)$	IG
2.60	1	(0, 1)	0.000000	7	(4, 3)	0.985228	0.862075	0.137925
2.80	3	(2, 1)	0.918296	5	(2, 3)	0.970951	0.951205	0.048795
2.95	4	(3, 1)	0.811278	4	(1, 3)	0.811278	0.811278	0.188722
3.10	6	(4, 2)	0.918296	2	(0, 2)	0.000000	0.688722	0.311278
3.30	7	(4, 3)	0.985228	1	(0, 1)	0.000000	0.862075	0.137925

· **petal_length(cm)**

θ	$ S_{\leq \theta} $	Left counts	H_L	$ S_{> \theta} $	Right counts	H_R	$H(S A, \theta)$	IG
4.05	1	(1, 0)	0.000000	7	(3, 4)	0.985228	0.862075	0.137925
4.45	2	(2, 0)	0.000000	6	(2, 4)	0.918296	0.688722	0.311278
4.90	3	(3, 0)	0.000000	5	(1, 4)	0.721928	0.451205	0.548795
5.30	4	(4, 0)	0.000000	4	(0, 4)	0.000000	0.000000	1.000000
5.55	5	(4, 1)	0.721928	3	(0, 3)	0.000000	0.451205	0.548795
5.70	6	(4, 2)	0.918296	2	(0, 2)	0.000000	0.688722	0.311278
5.85	7	(4, 3)	0.985228	1	(0, 1)	0.000000	0.862075	0.137925

· **petal_width(cm)"**

θ	$ S_{\leq \theta} $	Left counts	H_L	$ S_{> \theta} $	Right counts	H_R	$H(S A, \theta)$	IG
1.30	2	(2, 0)	0.000000	6	(2, 4)	0.918296	0.688722	0.311278
1.50	3	(3, 0)	0.000000	5	(1, 4)	0.721928	0.451205	0.548795
1.70	4	(4, 0)	0.000000	4	(0, 4)	0.000000	0.000000	1.000000
1.95	5	(4, 1)	0.721928	3	(0, 3)	0.000000	0.451205	0.548795
2.20	6	(4, 2)	0.918296	2	(0, 2)	0.000000	0.688722	0.311278
2.35	7	(4, 3)	0.985228	1	(0, 1)	0.000000	0.862075	0.137925

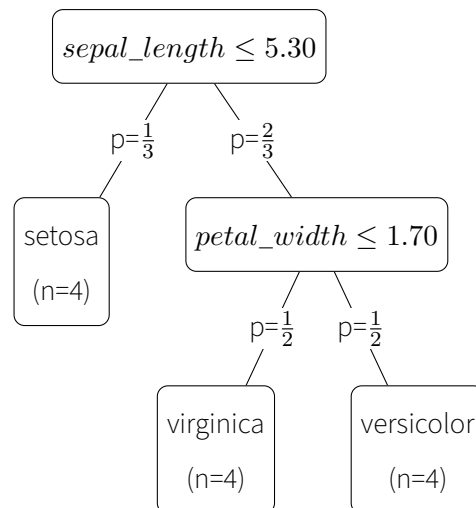
1.2.2.12 Final tree

On this child node we also end up with a few candidates with an equal IG score:

- $IG(S_{sepal_length > 5.30}, sepal_length \leq 6.20) = 1$
- $IG(S_{sepal_length > 5.30}, petal_length \leq 5.30) = 1$
- $IG(S_{sepal_length > 5.30}, petal_width \leq 1.70) = 1$

With the current training data anyone of these choices will result in a similar tree shape.

I'll make $IG(S_{sepal_length > 5.30}, petal_width \leq 1.70) = 1$ the node which results in this tree shape.



1.2.3 Gini Impurity

1.2.3.1 General formula

$$I_G(p) = \sum_{i=1}^J (p_i \sum_{k \neq i} p_k) = \sum_{i=1}^J p_i (1 - p_i) = \sum_{i=1}^J (p_i - p_i^2) = \sum_{i=1}^J p_i - \sum_{i=1}^J p_i^2 = 1 - \sum_{i=1}^J p_i^2$$

1.2.3.2 Finding the Root Node

When we're dealing with Gini Impurity, then it's all about finding the node with the lowest impurity score.

As a starting point we'll begin with finding the overall impurity of the training set by looking at how the target in the training set is divided. Also known as *Root Impurity*.

Looking at the target column (*class*) we can see that the possible outcomes are equally divided by $\frac{1}{3}$ which gives us the following calculation.

$$G_{root} = 1 - 3 \cdot \left(\frac{1}{3}\right)^2 = 1 - 3 \cdot \frac{1}{9} = 1 - \frac{1}{3} = \frac{2}{3} = 0.6667$$

1.2.3.3 Calculating Impurity

With this value we can now start to compare which feature will give the best Gini gain (reduce the impurity). This is done much in the same way as with entropy, but instead of looking for an increase in information gain, we're looking for a reduction of impurity.

First we establish the candidate thresholds which already has been covered in the entropy chapter. Then we'll do a weighted estimate of each threshold value with the following formulas.

$$G_{Left/Right} = 1 - \left(\frac{x_1}{x_1 + x_2 + x_3}\right)^2 - \left(\frac{x_2}{x_1 + x_2 + x_3}\right)^2 - \left(\frac{x_3}{x_1 + x_2 + x_3}\right)^2$$

$$Weighted\ child\ Gini : \quad G_{cond} = \frac{L_{x_1+x_2+x_3}}{n} \cdot G_L + \frac{R_{x_1+x_2+x_3}}{n} \cdot G_R$$

$$Gini\ gain : \quad G_{gain} = G_{root} - G_{cond}$$

Then the calculations begins.

– **sepal_length(cm)**

$$\theta \in \{4.5, 4.75, 5.3, 5.75, 5.9, 6.05, 6.2, 6.5, 6.75\}$$

Here is one example of the calculation that's being done for each θ in the feature.

$$\begin{aligned}\theta &= 4.5, \quad G_L \leq 4.5 \rightarrow (2, 0, 0), \quad G_R > 4.5 \rightarrow (2, 4, 4), \\ G_L &= 1 - \left(\frac{2}{2+0+0}\right)^2 - \left(\frac{0}{2+0+0}\right)^2 - \left(\frac{0}{2+0+0}\right)^2 = 0 \\ G_R &= 1 - \left(\frac{2}{2+4+4}\right)^2 - \left(\frac{4}{2+4+4}\right)^2 - \left(\frac{4}{2+4+4}\right)^2 = 0.64 \\ G_{cond} &= \frac{2}{12} \cdot 0 + \frac{10}{12} \cdot 0.64 = 0.53333 \\ G_{gain} &= 0.6667 - 0.53333 = 0.13337\end{aligned}$$

Here we can observe that $\theta = 4.5$ would reduce the impurity by **0.13337**, which would give us a *Gini Impurity* of $0.6667 - 0.13337 = \underline{0.53333}$ if we were to use this threshold as a root node for our tree.

We do the same calculations for each node to find the threshold with the best G_{gain} .

θ	$ n_{left} $	Left counts	G_L	$ n_{right} $	Right counts	G_R	G_{cond}	G_{gain}
4.50	2	(2, 0, 0)	0.000000	10	(2, 4, 4)	0.640000	0.533333	0.133333
4.75	3	(3, 0, 0)	0.000000	9	(1, 4, 4)	0.592593	0.444444	0.222222
5.30	4	(4, 0, 0)	0.000000	8	(0, 4, 4)	0.500000	0.333333	0.333333
5.75	5	(4, 1, 0)	0.320000	7	(0, 3, 4)	0.489796	0.419048	0.247619
5.90	6	(4, 2, 0)	0.444444	6	(0, 2, 4)	0.444444	0.444444	0.222222
6.05	7	(4, 3, 0)	0.489796	5	(0, 1, 4)	0.320000	0.419048	0.247619
6.20	8	(4, 4, 0)	0.500000	4	(0, 0, 4)	0.000000	0.333333	0.333333
6.50	9	(4, 4, 1)	0.592593	3	(0, 0, 3)	0.000000	0.444444	0.222222
6.75	10	(4, 4, 2)	0.640000	2	(0, 0, 2)	0.000000	0.533333	0.133333

– **sepal_width(cm)**

$$\theta \in \{2.6, 2.8, 2.95, 3.05, 3.15, 3.3\}$$

θ	$ n_{left} $	Left counts	G_L	$ n_{right} $	Right counts	G_R	G_{cond}	G_{gain}
2.60	1	(0, 0, 1)	0.000000	11	(4, 4, 3)	0.661157	0.606061	0.060606
2.80	3	(0, 2, 1)	0.444444	9	(4, 2, 3)	0.641975	0.592593	0.074074
2.95	5	(1, 3, 1)	0.560000	7	(3, 1, 3)	0.612245	0.590476	0.076190
3.05	7	(1, 4, 2)	0.571429	5	(3, 0, 2)	0.480000	0.533333	0.133333
3.15	8	(2, 4, 2)	0.625000	4	(2, 0, 2)	0.500000	0.583333	0.083333
3.30	11	(4, 4, 3)	0.661157	1	(0, 0, 1)	0.000000	0.606061	0.060606

– **petal_length(cm)**

$$\theta \in \{1.35, 1.45, 2.7, 4.05, 4.45, 4.9, 5.3, 5.55, 5.69, 5.85\}$$

θ	$ n_{left} $	Left counts	G_L	$ n_{right} $	Right counts	G_R	G_{cond}	G_{gain}
1.35	1	(1, 0, 0)	0.000000	11	(3, 4, 4)	0.661157	0.606061	0.060606
1.45	3	(3, 0, 0)	0.000000	9	(1, 4, 4)	0.592593	0.444444	0.222222
2.70	4	(4, 0, 0)	0.000000	8	(0, 4, 4)	0.500000	0.333333	0.333333
4.05	5	(4, 1, 0)	0.320000	7	(0, 3, 4)	0.489796	0.419048	0.247619
4.45	6	(4, 2, 0)	0.444444	6	(0, 2, 4)	0.444444	0.444444	0.222222
4.90	7	(4, 3, 0)	0.489796	5	(0, 1, 4)	0.320000	0.419048	0.247619
5.30	8	(4, 4, 0)	0.500000	4	(0, 0, 4)	0.000000	0.333333	0.333333
5.55	9	(4, 4, 1)	0.592593	3	(0, 0, 3)	0.000000	0.444444	0.222222
5.70	10	(4, 4, 2)	0.640000	2	(0, 0, 2)	0.000000	0.533333	0.133333
5.85	11	(4, 4, 3)	0.661157	1	(0, 0, 1)	0.000000	0.606061	0.060606

– **petal_width(cm)**

$$\theta \in \{0.15, 0.70, 1.30, 1.50, 1.70, 1.95, 2.20, 2.35\}$$

θ	$ n_{left} $	Left counts	G_L	$ n_{right} $	Right counts	G_R	G_{cond}	G_{gain}
0.15	1	(1, 0, 0)	0.000000	11	(3, 4, 4)	0.661157	0.606061	0.060606
0.70	4	(4, 0, 0)	0.000000	8	(0, 4, 4)	0.500000	0.333333	0.333333
1.30	6	(4, 2, 0)	0.444444	6	(0, 2, 4)	0.444444	0.444444	0.222222
1.50	7	(4, 3, 0)	0.489796	5	(0, 1, 4)	0.320000	0.419048	0.247619
1.70	8	(4, 4, 0)	0.500000	4	(0, 0, 4)	0.000000	0.333333	0.333333
1.95	9	(4, 4, 1)	0.592593	3	(0, 0, 3)	0.000000	0.444444	0.222222
2.20	10	(4, 4, 2)	0.640000	2	(0, 0, 2)	0.000000	0.533333	0.133333
2.35	11	(4, 4, 3)	0.661157	1	(0, 0, 1)	0.000000	0.606061	0.060606

1.2.3.4 Choosing the First Node**Summarizing results**

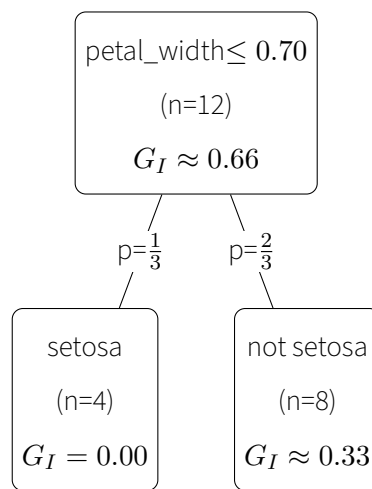
$$\cdot G_{gain}(sepal_length, 5.3) = 0.333333$$

$$\cdot G_{gain}(sepal_length, 6.2) = 0.333333$$

- $G_{gain}(\text{petal_length}, 2.70) = 0.333333$
- $G_{gain}(\text{petal_length}, 5.30) = 0.333333$
- $G_{gain}(\text{petal_width}, 0.70) = 0.333333$
- $G_{gain}(\text{petal_width}, 1.70) = 0.333333$

Here we also end up with several candidates which would reduce the impurity as much as possible. Here we take a choice at random and start to calculate the impurities based on the choice of root node for our tree.

For this round we'll make $G_{gain}(\text{petal_width}, 0.70) = 0.333333$ the root of our tree which gives us the following structure to build from.



1.2.3.5 Recompute Impurity with Binary Thresholds

We now repeat the process but exclude the rows which has been allocated correctly by the root node.

– **sepal_length(cm)**

θ	$ n_{left} $	Left counts	G_L	$ n_{right} $	Right counts	G_R	G_{cond}	G_{gain}
5.75	1	(1, 0)	0.000000	7	(3, 4)	0.489796	0.428571	0.071429
5.90	2	(2, 0)	0.000000	6	(2, 4)	0.444444	0.333333	0.166667
6.05	3	(3, 0)	0.000000	5	(1, 4)	0.320000	0.200000	0.300000
6.20	4	(4, 0)	0.000000	4	(0, 4)	0.000000	0.000000	0.500000
6.50	5	(4, 1)	0.320000	3	(0, 3)	0.000000	0.200000	0.300000
6.75	6	(4, 2)	0.444444	2	(0, 2)	0.000000	0.333333	0.166667

– **sepal_width(cm)**

θ	$ n_{left} $	Left counts	G_L	$ n_{right} $	Right counts	G_R	G_{cond}	G_{gain}
2.60	1	(0, 1)	0.000000	7	(4, 3)	0.489796	0.428571	0.071429
2.80	3	(2, 1)	0.444444	5	(2, 3)	0.480000	0.466667	0.033333
2.95	4	(3, 1)	0.375000	4	(1, 3)	0.375000	0.375000	0.125000
3.10	6	(4, 2)	0.444444	2	(0, 2)	0.000000	0.333333	0.166667
3.30	7	(4, 3)	0.489796	1	(0, 1)	0.000000	0.428571	0.071429

– **petal_length(cm)**

θ	$ n_{left} $	Left counts	G_L	$ n_{right} $	Right counts	G_R	G_{cond}	G_{gain}
4.05	1	(1, 0)	0.000000	7	(3, 4)	0.489796	0.428571	0.071429
4.45	2	(2, 0)	0.000000	6	(2, 4)	0.444444	0.333333	0.166667
4.90	3	(3, 0)	0.000000	5	(1, 4)	0.320000	0.200000	0.300000
5.30	4	(4, 0)	0.000000	4	(0, 4)	0.000000	0.000000	0.500000
5.55	5	(4, 1)	0.320000	3	(0, 3)	0.000000	0.200000	0.300000
5.70	6	(4, 2)	0.444444	2	(0, 2)	0.000000	0.333333	0.166667
5.85	7	(4, 3)	0.489796	1	(0, 1)	0.000000	0.428571	0.071429

– **petal_width(cm)**

θ	$ n_{left} $	Left counts	G_L	$ n_{right} $	Right counts	G_R	G_{cond}	G_{gain}
1.30	2	(2, 0)	0.000000	6	(2, 4)	0.444444	0.333333	0.166667
1.50	3	(3, 0)	0.000000	5	(1, 4)	0.320000	0.200000	0.300000
1.70	4	(4, 0)	0.000000	4	(0, 4)	0.000000	0.000000	0.500000
1.95	5	(4, 1)	0.320000	3	(0, 3)	0.000000	0.200000	0.300000
2.20	6	(4, 2)	0.444444	2	(0, 2)	0.000000	0.333333	0.166667
2.35	7	(4, 3)	0.489796	1	(0, 1)	0.000000	0.428571	0.071429

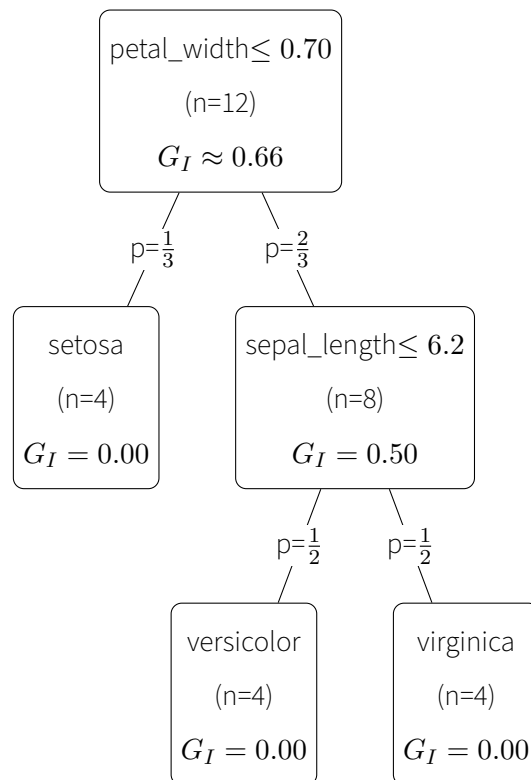
1.2.3.6 Best Candidates after Threshold Split

- $G_{gain}(sepal_length, 6.2) = 0.500000$
- $G_{gain}(petal_length, 5.30) = 0.500000$
- $G_{gain}(petal_width, 1.70) = 0.500000$

Here we end up with three candidates which has a 50/50 impurity. We chose one and structure the final tree.

1.2.3.7 petal_width ≤ 0.70 – Child Node

For this case we choose $\text{sepal_length} \leq 6.2$ as our child node, which gives us the final tree structure.

1.2.3.8 Final tree

Comparison of Classifiers for Breast Cancer Prediction

2.1 Task

The objective in this assignment is to create both a [decision tree classifier](#) and a [naive bayes classifier](#) for the [Breast Cancer Prediction dataset](#) ([names file](#), [UCI link](#) for description) and to compare their performance. What is the best algorithm for the dataset?

- Describe the dataset. What is the purpose of the dataset? What are the features, and what is the target? Write with your own words
- Perform necessary pre-processing of data. In particular, remove unusable columns.
- Split the dataset into suitable partitions using a random sampling technique. Defend your choice of split percentage(s) and sampling technique.
- You must use hyperparameter tuning to find the best possible classifiers. The choice of tuning parameters and their values and corresponding results should be discussed.
- The classifiers must be evaluating using accuracy, precision, recall and the f-measure. Discuss your results.

2.2 Solution

2.2.1 Description of the Dataset

This dataset is a collection of several test performed to analyse breast tissue for potential cancer cells (*features computed from digitized images of fine-needle aspirates (FNA)*). The provided data that "*describe characteristics of the cell nuclei present in the image*"^[1].

The data contains 569 instances of data collection, with 32 attributes whereas 30 of these are real-valued input features (ID, Diagnosis, + 30 features).

It has no missing feature in the whole dataset and the target is weighted 63%/37% with 357 benign and 212 malignant rows.

The first usage of this data was used in a medicinal article under the name "*Nuclear feature extraction for breast tumor diagnosis*", which I suspect was an article intended within the medical field to spread knowledge about how cancerous cells formed in breast to further improve diagnostic on malignant cancer cells.

2.2.1.1 Purpose

It was donated to *UC Irvine* in 1995 as statistical data with the purpose of training data for developing machine learning algorithms, which then could be used to further increase the probability of discovering potential malignant cell development.

According to the appended documentation to this data, there has been developed several models to do these predictions with one of the top models was a *classifier* with an accuracy up to 97.5%.

2.2.1.2 Features

The dataset is built up with the following features:

- radius(1,2,3)
- texture(1,2,3)
- perimeter(1,2,3)
- area(1,2,3)
- smoothness(1,2,3)
- compactness(1,2,3)
- concavity(1,2,3)
- concave_points(1,2,3)
- symmetry(1,2,3)
- fractal_dimension(1,2,3)

Each feature has three columns which represents different data collections for each image.

- 1) Mean — the average value across the nuclei in the image
- 2) SE (standard error) — a variability estimate of that feature across the nuclei
- 3) Worst — the “largest” value, defined as the mean of the three largest nuclei values

2.2.1.3 Target

The target of this dataset is labelled "*Diagnosis*" and contains information that states if the cell nuclei were considered benign (of no danger to health) or malignant (tending to metastasize), which makes this dataset ideal as training material for models as we end up with a binary classification based on true/false statements.

2.2.2 Models

The rest of this task is to be found in the appended notebook named "*wdbc.ipynb*".

Regression

3.1 Task

3.1.1 Comparison of Regressors for Real Estate Valuation

In this assignment, the objective is to create both a [decision tree regressor](#) and a [linear regression](#) model for the [Real Estate Valuation dataset](#) and to compare their performance. What is the best algorithm for the dataset?

- Describe the dataset. What is the purpose of the dataset? What are the features, and what is the target? Write with your own words
- Perform necessary pre-processing of data. In particular, remove unusable columns.
- Split the dataset into suitable partitions using a random sampling technique. Defend your choice of split percentage(s) and sampling technique.
- You must use hyperparameter tuning to find the best possible regressor. The choice of tuning parameters and their values and corresponding results should be discussed.
- The regressors must be evaluated using r^2 , absolute error, and mean squared error. Discuss your results.

3.2 Solution

3.2.1 Description of the Dataset

So this file didn't have a reference document like the *wdbc* dataset. But after some digging and comparing of data, I managed to deduct that this data derives from a [Real Estate Valuation](#) in Taiwan[2].

This dataset contains historical collections of real estate evaluations from *New Taipei City* in the period 2012-2013.

The page also mentions that this dataset has earlier been divided into 2/3 train and 1/3 test, but no information about the result of this partitioning.

- **Data contains:**
 - 414 instances
 - No missing values
 - 6 Features

- 1 Target
- 2 counting columns with one missing label.
(Most likely an formatting error in the data)

3.2.1.1 Purpose

I'm sure that there are several potentials for this type of data collections.

The first thing that comes to mind is a statistical analysis of area pricing over time, to try and estimate future price fluctuation in the given areas. This could then be used to make prediction on when house prices are expected to rise and at which point it would be considered wise to sell or wait and what price offers one potentially could expect to get in certain periods.

Ideally one would quite a lot more data then what this set contains to make more accurate predictions. I imagine that something in the range of 10+ years of this data would drastically improve the predictions of the trained model.

3.2.1.2 Features

X1 – The transaction date (for example, 2013.250=2013 March, 2013.500=2013 June, etc.)

X2 – The house age (unit: year)

X3 – The distance to the nearest MRT station (unit: meter)

X4 – The number of convenience stores in the living circle on foot (integer)

X5 – The geographic coordinate, latitude. (unit: degree)

X6 – The geographic coordinate, longitude. (unit: degree)

3.2.1.3 Target

Y – House price of unit area.

The value structure here is quite unfamiliar to me, but I do think that a target instance represents the estimated pricing per ping ($1\text{ping} \approx 3.3\text{m}^2$) where 1 value equals 10'000 New Taiwan Dollar(NTD)[3]. So if $Y = 37.9$ means 379'000NTD($\approx 124.50\text{NOK}$) per 3.3m^2 .

3.2.2 Models

The rest of this task is to be found in the appended notebook named "*realEstate.ipynb*".

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